

Business Requirement Document (BRD)

Project Title

E-Commerce Order Management Database System

1. Business Objective

The objective of this project is to develop a centralized database system for an E-Commerce Order Management System that efficiently manages customers, products, suppliers, orders, payments, shipments, and customer reviews. The system aims to improve operational efficiency, reduce data redundancy, maintain data consistency, support faster order processing, and generate meaningful business reports for informed decision-making.

2. Business Goals

The proposed system aims to achieve the following goals:

- Improve customer shopping experience.
 - Maintain accurate customer and product records.
 - Reduce manual data processing.
 - Automate order management.
 - Improve inventory tracking.
 - Secure payment information.
 - Monitor shipment status.
 - Store customer product reviews.
 - Generate business reports.
 - Improve overall business productivity.
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3. Business Requirements

The proposed system should satisfy the following business requirements:

Customer Management

- Customer registration

- Customer profile management
- Secure login
- Address management

Product Management

- Product catalog
- Product categorization
- Product pricing
- Stock quantity management

Supplier Management

- Supplier information management
- Contact information maintenance

Order Management

- Order placement
- Order tracking
- Order cancellation
- Order history

Payment Management

- Multiple payment methods
- Payment status tracking
- Payment history

Shipment Management

- Shipment scheduling
- Delivery tracking
- Delivery status updates

Review Management

- Product ratings
- Customer comments
- Review history

4. Business Scope

The project includes the following features:

In Scope

- Customer Registration
- Product Management
- Category Management
- Supplier Management
- Order Processing
- Payment Processing
- Shipment Tracking
- Customer Reviews
- Business Report Generation

Out of Scope

- AI Product Recommendations
- Live Chat Support
- Mobile Application
- Coupon Management
- Loyalty Rewards
- Social Media Integration

5. Stakeholder Analysis

Stakeholder	Responsibility
Customer	Purchase products and provide reviews
Administrator	Manage complete system
Supplier	Supply products
Warehouse Staff	Manage inventory

Stakeholder	Responsibility
Delivery Staff	Deliver customer orders
Finance Department	Verify payments
System Administrator	Maintain database and security

6. Business Rules

The following business rules must be followed:

- Every customer must have a unique Customer ID.
 - Email addresses must be unique.
 - Every product belongs to one category.
 - Stock quantity cannot be negative.
 - Orders must contain at least one product.
 - Payment must be completed before shipment.
 - Shipment is created only after payment confirmation.
 - Ratings are allowed only from 1 to 5.
 - Reviews can only be submitted by registered customers.
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7. Functional Requirements

The system shall:

- Register new customers.
- Maintain customer information.
- Add, update, and delete products.
- Manage product categories.
- Maintain supplier details.
- Create customer orders.
- Process payments.
- Generate invoices.

- Track shipments.
 - Store customer reviews.
 - Generate reports.
 - Search customer records.
 - Search product information.
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8. Non-Functional Requirements

The system should provide:

Performance

- Fast response time.

Security

- Secure user authentication.
- Password encryption.
- Role-based access.

Reliability

- Accurate data storage.
- Data consistency.

Availability

- System available 24×7.

Scalability

- Support future business growth.

Maintainability

- Easy software maintenance.

Backup and Recovery

- Automatic data backup.
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9. Assumptions

- Customers have internet access.

- Products are available in inventory.
 - Payments are processed successfully.
 - Delivery services are operational.
 - The database server is continuously available.
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10. Constraints

- Only the given entities are used.
 - No modification of entity attributes.
 - Database follows relational design.
 - Internet connection is required.
 - Users require authentication.
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11. Expected Business Benefits

Implementation of this system will provide:

- Reduced paperwork.
 - Faster order processing.
 - Improved inventory management.
 - Better customer satisfaction.
 - Increased sales efficiency.
 - Better business reporting.
 - Reduced data redundancy.
 - Improved decision-making.
 - Centralized database management.
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12. Conclusion

The proposed E-Commerce Order Management Database System provides a centralized and efficient solution for managing online business operations. By integrating customer management, product management, order processing, payment tracking, shipment monitoring, and customer reviews into a single database system,

the organization can improve operational efficiency, enhance customer satisfaction, and support informed business decision-making. This Business Requirement Document serves as the foundation for the subsequent stages of database design, implementation, testing, and deployment.